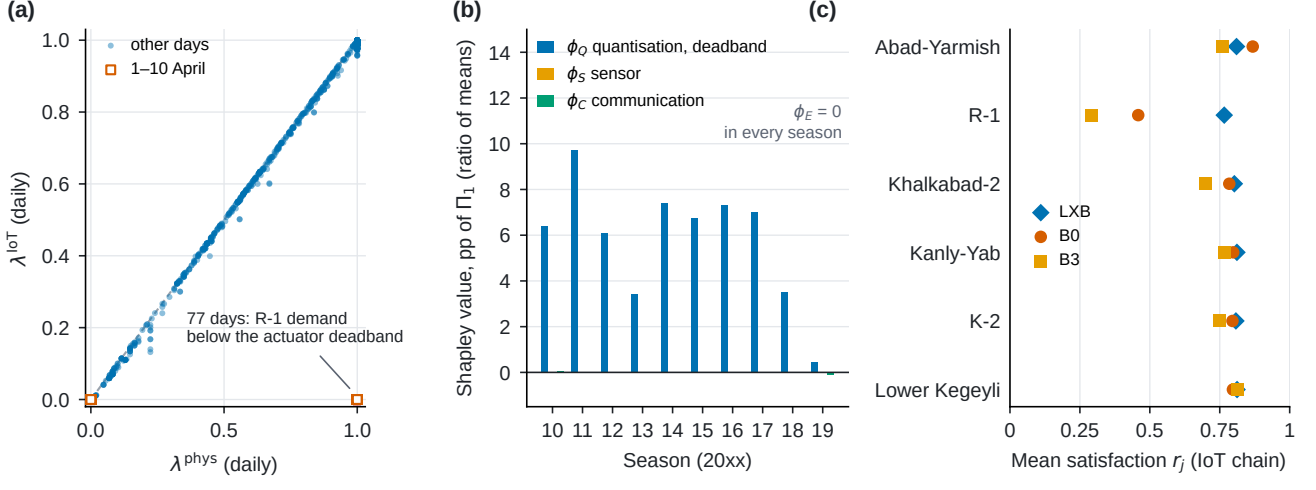

(a)
other days
1–10 April
77 days: R-1 demand
below the actuator deadband
$\lambda^{\mathrm{IoT}}$ (daily)
$\lambda^{\mathrm{phys}}$ (daily)
(b)
$\phi_Q$ quantisation, deadband
$\phi_S$ sensor
$\phi_C$ communication
$\phi_E = 0$
in every season
Shapley value, pp of $\Pi_1$ (ratio of means)
Season (20xx)
(c)
Abad-Yarmish
R-1
Khalkabad-2
Kanly-Yab
K-2
Lower Kegeyli
LXB
B0
B3
Mean satisfaction $r_j$ (IoT chain)